Bugs and Insect Species Data Sources

- Global Biodiversity Information Facility (GBIF) – A worldwide biodiversity data aggregator that includes millions of species occurrence records. GBIF can be filtered for Kansas to retrieve insect observations (from museum specimens, field surveys, and citizen science) across the state .
    - Coverage: All of Kansas (and globally), with records spanning the 1800s to present (dates and locations for each occurrence).
    - Access: Open API and bulk downloads (CSV/DwC) via GBIF's portal. Users can query by taxon (e.g. insect orders) and geography (state or coordinates).
    - License: Data are open-access; most records carry Creative Commons designations (CC0 or CC BY) permitting unrestricted use with attribution .
    - Potential Use: Enables mapping of insect distributions and diversity over time. For the Kansas Frontier Matrix, GBIF data can help model species richness, track ecological changes, or train AI to recognize habitat associations of insects.
- iNaturalist Observations – A citizen-science platform with thousands of crowd-sourced insect sightings in Kansas. Participants upload geo-tagged insect photos which are identified by the community.
    - Coverage: Statewide, especially in populated or surveyed areas; data mainly from the last ~10–15 years (continuous updates). Includes date, location, and species for each observation (from butterflies to beetles).
    - Access: REST API and project downloads. Data (CSV/JSON) can be pulled by taxonomic group and region. An account is required for full dataset export.
    - License: By default, iNaturalist observation data are shared under CC BY-NC (allowing use with attribution for non-commercial purposes) . Many contributors opt for even more open licenses, and research-quality records are exported to GBIF.
    - Potential Use: iNaturalist provides real-time time-series of insect occurrences and seasonal activity. These data can enrich models of insect phenology, support pollinator studies, or validate historical species range maps with contemporary sightings.

Temperatures and Climate Data

- NOAA Historical Climate Data (NCEI) – The National Centers for Environmental Information host extensive weather and climate records for Kansas. This includes daily and monthly measurements from station networks (e.g. COOP, airport stations) covering temperature, precipitation, snowfall, etc. .

    - Coverage: Numerous stations across Kansas from the late 1800s to current day, plus statewide aggregated climate normals. Data spans historical periods (19th–20th century) to the present, enabling long-term climate trend analysis.

    - Access: Free download via Climate Data Online (CSV files or JSON via API) . Users can query by station or county; GIS-ready datasets (e.g. gridded climate normals) are also available.

    - License: U.S. government data – public domain (free to use without restriction). NOAA explicitly provides these records for public use .

    - Potential Use: These datasets allow reconstruction of Kansas's historical climate – e.g. tracking 20th-century temperature trends or extreme drought periods. In the Frontier Matrix, they could drive environmental context for agricultural yields, human settlement patterns (e.g. Dust Bowl migrations), or feed AI models predicting future climate scenarios.

- Kansas Mesonet – A network of local weather stations operated by Kansas State University, offering high-resolution climate observations. The Mesonet monitors temperature, rainfall, wind, and soil moisture at 60+ stations across Kansas.

    - Coverage: Modern coverage (since ~1980s–1990s) at a fine spatial scale, with observations every 5 minutes to hourly. Stations are distributed statewide (agricultural experiment fields, etc.), capturing microclimate variations.

    - Access: Open RESTful API delivering data in CSV format . Users can query specific stations and date ranges; up to 3,000 records per request (approx. 8 years of daily data) . The Mesonet site also provides a URL builder for custom queries.

    - License: Free public use with attribution – "Data accessed through mesonet.k-state.edu is free for public use and download" (with request to cite the source) . Data is preliminary (quality-controlled over time).

    - Potential Use: Ideal for fine-grained recent climate analysis – e.g. comparing microclimates between prairie and irrigated cropland, or assessing soil moisture

trends. The Kansas Frontier Matrix can integrate Mesonet data to model localized impacts of weather on crop yields, drought conditions, or even tie climate anomalies to historical events (like floods or heatwaves that influenced settlement).

Wildlife and Animal Population Data

- eBird (Cornell Lab of Ornithology) – A massive open dataset of bird observations, which includes tens of thousands of records from Kansas. Birdwatchers submit checklists with species seen, count, date, and location.
    - Coverage: All Kansas counties, with data from ~2000 to present (and some historical checklists). Provides seasonal and spatial coverage for 450+ bird species in Kansas, often at daily resolution. Trend metrics (population changes) are updated annually .
    - Access: eBird API (for recent sightings) and bulk data downloads. The eBird Basic Dataset (EBD) can be requested (free) which contains all observations; state-level summary products (relative abundance rasters, trend maps) are also available .
    - License: Free for non-commercial research and education use. Users must agree to data terms (eBird requires registration and acceptance of a data use agreement) . Data must be cited and cannot be redistributed in raw form. (eBird also contributes records to GBIF under CC licenses.)
    - Potential Use: eBird provides a rich time series of wildlife data – for example, mapping the decline of prairie-chickens or the spread of invasive bird species in Kansas. These observations can calibrate species distribution models or serve as a proxy for ecosystem health. In the Frontier Matrix, eBird data could animate changes in biodiversity, inform conservation scenario modeling, or augment AI analyses of habitat change.
- Kansas GAP Species Distribution Maps – The Kansas Gap Analysis Project (GAP) produced predicted habitat distribution datasets for the state's wildlife . These GIS layers model the suitable range of various species (mammals, birds, reptiles, etc.) based on land cover and known occurrence records (circa 2000).
    - Coverage: Statewide coverage for dozens of species. For example, GAP provides raster maps (~30–90 m resolution) of potential habitat for all breeding birds, mammals, amphibians present in Kansas as of 2000 . Temporal coverage is a single baseline (no annual change, as it's a habitat model snapshot around

2000).

- Access: Downloadable GIS files (ArcINFO grids and shapefiles) via the Kansas Data Access & Support Center. Metadata and links are available on the Kansas GIS portal . No update is planned (static dataset), but it is in the public domain for use.

- License: Public domain (U.S. Geological Survey funded). The data are freely usable; citation of the Kansas GAP project (2002) is suggested.

- Potential Use: Provides a pre-settlement or early-2000s baseline of where wildlife should occur based on habitat. This is invaluable for historical ecological modeling – e.g., comparing predicted bison or pronghorn ranges with 19th-century accounts. In the Frontier Matrix, GAP maps can be layered with land-use change to identify “gaps” in conservation or to guide AI in understanding historical species distribution versus current observations (from eBird/GBIF).

Census and Demographic Data

- U.S. Census Bureau – Decennial Census & ACS – Authoritative population and housing data for Kansas from the 19th century to today. Decennial censuses (every 10 years) provide official population counts and demographics for 1860, 1870, … 2020, while the American Community Survey (annual/5-year rolling) provides current estimates.

    - Coverage: Historical and current. Decennial census data exist for Kansas as far back as 1860 (pre-statehood territorial census data for 1850s also available) through 2020, at geographies from state, county, down to tracts/block groups (in later years). The ACS covers roughly 2005–present with detailed socio-economic variables.

    - Access: Modern data (1990s–present) can be queried via the Census API or downloaded from data.census.gov in CSV. Historical data (1790–1980s) are accessible via archives or specialized portals (see NHGIS below). Boundary files (shapefiles for counties, tracts, etc.) are also published by the Census.

    - License: U.S. Census data are public domain (no license restrictions) – “Open Government Data”. The Census API terms mainly require attribution and avoid misuse, but the data itself is free to reuse.

    - Potential Use: Demographic data is fundamental for historical modeling. It enables analysis of population growth, density changes on the frontier, urbanization patterns (e.g. the rise/fall of railroad towns), or mapping ethnic

composition over time. In the Kansas Frontier Matrix, census stats can drive visualizations of settlement expansion, power AI predictions of future demographic shifts, or correlate population with environmental factors (like how drought influenced rural depopulation).

- IPUMS NHGIS (National Historical GIS) – A curated repository of historical census statistics and GIS boundary files . NHGIS provides easy access to aggregated census data for every decade, along with consistent digital maps for historical county and tract boundaries.

    - Coverage: 1790 through present for decennial census data . For Kansas, NHGIS includes territorial census data, all decennial counts, and even historical agriculture and economic census tables. Geographically, one can retrieve data at state, county, or even enumeration district levels (depending on year). Boundaries reflect the changing county lines (e.g. the many county splits in Kansas's 19th-century).

    - Access: Free with registration. Users select desired years, geographic levels, and data tables on the NHGIS website, then download a custom zip with CSV data and matching shapefiles. NHGIS also offers time-series tables that link variables across decades.

    - License: Free to use for any purpose, but users must agree to cite the source. The data originates from public domain census records; IPUMS asks for attribution for the compilation. There are no commercial restrictions.

    - Potential Use: NHGIS simplifies integrating historical census data into GIS. For example, one can map population density on the Kansas frontier in 1870 versus 1970 using authentic county outlines. It's invaluable for creating longitudinal models – the Frontier Matrix can use it to animate demographic change, train AI on how socio-economic variables (like rural farm populations) shifted over 150+ years, or to contextualize other datasets (e.g. overlaying climate or land use with population change).

Agriculture and Land Use Data

- USDA NASS QuickStats (Crop and Yield Data) – The National Agricultural Statistics Service's open database of U.S. agricultural data. QuickStats contains county-level and state-level statistics on crop acreage, yields, production, livestock numbers, economics, etc., from both annual surveys and the 5-year Census of Agriculture .

- Coverage: Kansas statewide and by county, with time series for many variables. Data spans the late 20th century to present (generally from ~1980s onward for surveys; 1997, 2002, 2007, 2012, 2017 Ag Censuses included). For example, one can retrieve the wheat yield per acre for every Kansas county for each year, or the number of farms by size class.

- Access: QuickStats API (JSON/CSV responses) and an online query tool . The API allows filtering by commodity, geographic area, year, etc. Bulk downloads of certain datasets (like ag census) are also on data.gov.

- License: Creative Commons CC0 (Public Domain) – NASS designates QuickStats data as open for public use without restriction .

- Potential Use: This is crucial for modeling agricultural trends. In the Frontier Matrix, NASS data can be used to plot the rise and fall of different crops (e.g. corn vs. sorghum acreage over decades), analyze yields relative to climate events, or train AI models to predict future crop outputs. It also helps tie historical events (like the Dust Bowl) to hard data (crop failures, livestock losses).

- USDA Cropland Data Layer (CDL) – An annual land cover raster that maps crop types across every field in Kansas . The CDL is produced from satellite imagery each year, identifying which crops (corn, wheat, soybeans, etc.) or land cover (forest, grass, urban) are present on each 30×30 m pixel.

    - Coverage: Complete statewide coverage. Available yearly from 2008 to present for Kansas (prior to 2008, the program covered fewer states) . Each year's layer is a georeferenced GeoTIFF, enabling time-series analysis of land use change.

    - Access: Downloadable via the CropScape web portal or directly from data.gov (as GeoTIFF per state per year) . Also accessible through a web map service (WMS) for GIS software . File size is significant but manageable (~50–100 MB per state-year).

    - License: Public domain (CC0) – NASS publishes the CDL as open data . No use restrictions; they encourage citation.

    - Potential Use: The CDL provides a richly detailed view of agricultural land use over time. For the Frontier Matrix, this enables mapping the expansion or contraction of certain crops (like the westward march of soybean cultivation, or areas converted from wheat to sorghum due to climate). It's also valuable for environmental modeling – e.g. relating crop rotations to soil health or carbon stock. AI applications might include classifying land cover changes or forecasting crop distribution under future scenarios.

Migration and Immigration Data

- Census Migration & Immigration Statistics – The U.S. Census Bureau tracks both domestic migration and international immigration characteristics which can be focused on Kansas. Decennial censuses in historical periods asked for place of birth (revealing how many Kansans were born abroad or in other states), and modern American Community Survey data report foreign-born populations, year of arrival, and mobility (moves in the past year).

    - Coverage: At the state and county level, one can obtain net migration figures and immigrant demographics. Historical example: the 1900 Census reported the number of Kansas residents born in Germany or Sweden (capturing immigrant communities). Contemporary data (ACS) detail, say, how many people moved into a given Kansas county in the last year, and from where (other state or abroad).

    - Access: Decennial data is available via NHGIS or IPUMS (for historical place-of-birth tables). ACS data (e.g. 5-year estimates) are accessible through the Census API – for instance, tables showing migration flows or nativity by county. The Census Bureau also occasionally releases specialized migration flow reports and tools on their website.

    - License: Public domain. As with all Census data, these stats are open-access for use and redistribution.

    - Potential Use: These data illuminate settlement patterns and population turnover. In a historical context, one could map the influx of settlers by origin (showing which Kansas counties had many residents from, say, Ohio or Germany in 1880). For modern times, one can track urbanization or youth out-migration from rural counties. Such patterns are vital for the Frontier Matrix to simulate how population shifts respond to economic booms, environmental pressures (e.g. the 1930s exodus), or policy changes. They also aid AI in correlating migration with factors like climate or crop failure.

- IRS Migration Data (County-to-County Flows) – An underutilized dataset from the IRS Statistics of Income division that captures yearly population movement via tax return address changes. It provides granular data on how many households and people moved into and out of each county (and state), along with their origin/destination .

    - Coverage: Nationwide, with specific files for Kansas. Annual data from 1991 through 2022 are available . For each Kansas county, one can see how many taxpayers left for each other state or county and how many arrived (inflows/outflows), as well as the aggregate Adjusted Gross Income of those

movers . This offers a quantitative view of migration trends in the late 20th and early 21st century.

- Access: Downloadable Excel/CSV files by year from the IRS website (the IRS provides state-level spreadsheets listing all county pairs) . No API, but data are structured and machine-readable. For earlier years (1990s), data come as zip files containing all states .

- License: As a U.S. government dataset, it's in the public domain. IRS makes it available for public use; there is no personal info (only aggregated counts).

- Potential Use: The IRS migration files are excellent for analyzing county-level migration patterns – e.g. which rural Kansas counties are losing population to urban areas, or net gains due to inbound retirees. They can complement Census estimates by providing year-by-year flow dynamics. In the Frontier Matrix, these could feed models of economic or climate-driven migration (for instance, out-migration from drought-hit counties) and help evaluate policy impacts (did a new interstate or industry park alter migration?). It's a rich dataset to train AI on socio-economic drivers of movement within Kansas.

Land Ownership and Land Patents

- BLM General Land Office (GLO) Records – A comprehensive archive of federal land patents and survey records for Kansas and other public-land states. This includes Homestead Act patents, railroad grants, military bounty land warrants, etc., documenting the transfer of land from the federal government to individuals or organizations .

    - Coverage: Historical (19th and early 20th century). Kansas, being a public domain state, has GLO patent records from the 1850s (Territorial Kansas surveys) through the early 1900s. Over 160,000 Kansas land patent records are available, each with the patentee name, date, legal land description (Township-Range-Section), and often an image of the original patent document.

    - Access: The GLO online database allows searching by person, location, or patent type. Results include digitized images (PDF) of original patents and metadata (which can be downloaded as XML). There is also an interactive map for patented land parcels. Bulk data download isn't directly provided on the site, but third-party compilations or scraping can retrieve large datasets. Coordinate data can be derived by linking to the Public Land Survey System grid.

    - License: Public domain. These are historic government records and free to use. The Bureau of Land Management encourages using the site for research and

provides the data openly (with the caveat that not every single record may be online) .

- Potential Use: GLO records are key to reconstructing settlement. They enable mapping original homesteads – for example, identifying when and where land was first claimed under the Homestead Act in western Kansas. In the Frontier Matrix, this could animate land acquisition over time, link family histories to specific plots, or feed models about how quickly the frontier filled up. Moreover, by intersecting patent maps with soil or rainfall data, one could analyze outcomes (did homesteads on marginal land get abandoned more often?). This data brings an important human dimension: actual names and dates of Kansas's frontier land owners.

- Public Land Survey System (PLSS) Grid – Geospatial data delineating the township, range, and section lines across Kansas. The PLSS is the land subdivision system used to describe locations of land parcels in Kansas (e.g., "T15S R21W Section 14"). Digital PLSS shapefiles provide the boundaries of every section (roughly one-mile squares) and their identifiers.

  - Coverage: Statewide, static. The PLSS was established in Kansas starting in the mid-19th century; the dataset represents the surveyed grid that covers virtually all of Kansas (except some river irregularities).

  - Access: Available through BLM's Geospatial Hub and Kansas GIS repositories. The BLM provides a national PLSS polygon layer (which can be filtered to Kansas) and often state-specific downloads. Data formats include Esri Shapefile and Geodatabase. Each section polygon has attributes for township, range, section number, meridian, etc.

  - License: Public domain (U.S. Government data). No license restrictions.

  - Potential Use: The PLSS grid is the spatial "key" for integrating historic land records. By overlaying GLO patent data on the PLSS, one can create maps of homesteaded sections. In a broader sense, the PLSS also helps standardize other data (e.g., soil and water datasets are sometimes indexed by township-range as well). For the Frontier Matrix, having the PLSS allows a user to query data by legal land description – important for historians and genealogists. It also provides a base layer to visualize how Kansas was partitioned and settled section by section.

Crime and Justice Data

- FBI Uniform Crime Reports / NIBRS Data – The FBI's national crime statistics programs compile data for Kansas from local law enforcement agencies. This includes historical Uniform Crime Reporting (UCR) summary data (index crimes, arrests, etc.) and the modern National Incident-Based Reporting System (NIBRS) which has granular incident-level data.

    - Coverage: Statewide and city/county level crime data from the mid-20th century to present. Summary UCR data (violent/property crime rates) exist annually back to the 1960s for Kansas. NIBRS incident data is available in detail from the 1990s/2000s onward (with Kansas law enforcement now fully NIBRS compliant). The data cover offense counts, clearance rates, arrests, and more.

    - Access: The FBI's Crime Data Explorer (CDE) provides an open API to query crime stats . One can retrieve data by state, year, offense type, etc., in JSON or CSV. Bulk datasets (e.g. entire NIBRS files) can be downloaded from FBI or the National Archive of Criminal Justice Data. Additionally, the Kansas Bureau of Investigation publishes annual Crime Index Reports (in PDF/Excel) which can be used for local details.

    - License: Public domain (U.S. Dept. of Justice data). The FBI Crime API is explicitly marked as public data (no copyright) . Users are free to reuse and share the statistics.

    - Potential Use: Crime data can be woven into the socio-cultural narrative of Kansas. Historically, one could examine crime rates during boom vs. bust periods (did the Dust Bowl or 1980s farm crisis coincide with rising crime?). Spatially, data could highlight urban-rural differences or justice system resource needs. For the Frontier Matrix, crime statistics add a layer of social context – useful for AI models that might predict community well-being or for overlaying with demographic and economic shifts. They also enable civic engagement apps, e.g. showing long-term trends in public safety across Kansas communities.

Archaeological and Historical Sites Data

- Kansas Historic Resources Inventory (KHRI) – A database of historic and archaeological sites surveyed in Kansas, maintained by the State Historic Preservation Office. It includes information on buildings, structures, objects, and some archaeological sites that have been formally recorded . Each entry typically has location coordinates, a description, and notes on significance.

    - Coverage: Thousands of historic properties across Kansas. The focus is on built environment (e.g. historic schools, churches, homesteads) and known

archaeological sites (especially those eligible for registers). Temporal coverage spans indigenous archaeological sites (pre-1850s) to structures from the more recent past (mid-20th century) that have historic value.

  - Access: Web interface with search and map (KHRI Online) . Non-registered users can search and view all approved records . Data includes basic metadata and often photographs or survey forms. For GIS usage, data can be requested or exported for public sites. (Exact coordinates of sensitive archaeological sites may be generalized to protect them .)

  - License: The data is publicly accessible for education and reference . As a state government resource, it's effectively open, though users must refrain from using it to disturb sites (Kansas law exempts precise archeological locations from open-records requests for protection). In summary: free to use with proper attribution, but some location details are restricted.

  - Potential Use: KHRI is a treasure trove for historical mapping. It can show where old forts were, trails, Native American village sites, or early settler structures. In the Frontier Matrix, these points of interest allow layering cultural heritage onto environmental and demographic data – for instance, visualizing how early trails and trading posts align with water sources or how settlement progressed county by county. AI could use the textual descriptions in KHRI for natural language processing, extracting common themes of frontier architecture or settlement challenges.

- National Register of Historic Places (NRHP) GIS Data – A nationwide open dataset of historic places listed on the National Register, including many in Kansas. The National Park Service provides spatial data (points/polygons) for NRHP listings , which range from well-known landmarks to archaeological districts.

  - Coverage: Over 1,600 NRHP listings in Kansas (as of recent counts), from Dodge City's historic district to Native American archaeological sites. Includes site name, location (lat/long), and listing information. This is historical-cultural data spanning all periods of Kansas history. Updated as new sites are listed.

  - Access: Downloadable from NPS's Cultural Resources GIS portal or via data.gov. The data are provided as a feature service and shapefiles, with attributes like reference number, name, address, and area. Because not all NRHP sites (especially archaeological ones) have precise coordinates publicly shown, some may be generalized points.

  - License: Public domain (federal government publication) . The NPS data is explicitly "public, non-restricted data" for NRHP locations, meant for broad use in

mapping and analysis.

- Potential Use: NRHP data flags sites of official historical significance. This can be used to ensure the Frontier Matrix includes cultural landmarks in its model – for example, identifying counties rich in heritage resources or using it as a layer in predictive modeling (perhaps an AI that predicts which sites might deserve future preservation based on patterns). It also aids story-telling: one can click on a county and immediately see notable historic sites or dig into narratives of an old frontier fort, integrating story maps with data.

Soil and Vegetation Data

- NRCS Soil Survey (SSURGO Database) – Detailed soil maps for Kansas, produced by the USDA Natural Resources Conservation Service. SSURGO contains geospatial soil polygon maps and extensive attributes (soil type, texture, chemistry, capability class, etc.) for each mapping unit . Essentially, it's the digital form of county soil surveys, representing over a century of soil data collection.

  - Coverage: Statewide at fine resolution (typically 1:12,000 scale). SSURGO is organized by soil survey areas (often by county) – all Kansas counties have complete coverage. Temporal: the data reflects surveys done mostly 1960s–2000s, but soil properties are relatively stable; updates occur as new surveys are completed or data refined.

  - Access: Available through the Web Soil Survey and for download via NRCS's Soil Data Access portal. One can download GIS shapefiles or a geodatabase for each county or use the Soil Data Access API for querying specific properties. NRCS also offers a gridded SSURGO (gSSURGO) which compiles the data statewide in raster form. Kansas's GIS portal provides ready-to-use SSURGO layer packages as well .

  - License: Public domain. Soil surveys are government data and freely available. NRCS even notes the SSURGO portal is "license-free, open-source" for access .

  - Potential Use: Soils are fundamental for environmental and agricultural modeling. By integrating SSURGO, the Frontier Matrix can, for instance, correlate crop yields with soil types (explaining why certain areas grew wheat vs. corn), or model erosion and land degradation over time. Soil data can inform AI models predicting suitability for crops or forecasting how past farming practices impacted land (e.g. areas prone to Dust Bowl erosion). It also enriches historical reconstructions – knowing soil quality helps explain settlement patterns

(homesteaders avoided certain sandy soils, etc.).

- Land Cover and Vegetation – NLCD and Historical Data – Comprehensive land cover information for Kansas is available from the National Land Cover Database (NLCD), as well as historical vegetation maps. NLCD is a Landsat-derived product classifying each 30m cell as agriculture, forest, grassland, urban, etc., and is updated regularly . In addition, Kansas-specific efforts (e.g. Kansas Biological Survey's 2017–2018 land cover map or the Kansas GAP land cover from the 1990s) provide snapshots of vegetation.

    - Coverage: NLCD: Nationwide including Kansas, with products for 2001, 2006, 2011, 2016, and now produced annually (latest 2019/2020) . Resolution 30m. Classes include cropland, various forest types, grassland, wetland, developed, etc. Historic maps: A 1992 Kansas land cover (from GAP) and modern high-resolution (10m) maps from satellite imagery are available. These cover the entire state as well.

    - Access: NLCD can be downloaded free from the MRLC website or data.gov, as GeoTIFF grids. Annual NLCD layers (for recent decades) are also accessible. Kansas-specific land cover datasets can be obtained from the Kansas Applied Remote Sensing Program (KARS) or DASC as GeoJSON/GeoTIFF .

    - License: Public domain for NLCD (created by USGS; data is explicitly stated as nationwide public domain land cover information). State-derived products from KARS/KBS are generally open for public use as well (often funded by state/federal grants).

    - Potential Use: Land cover data allows the Frontier Matrix to incorporate the natural landscape context. For example, one could track the decline of prairie over decades or identify regions of forest regrowth. NLCD's consistent time-series supports analyzing land cover change (like urban sprawl around Kansas City or shifts in rangeland vs. cropland). In modeling, this is vital for habitat and environmental analyses: an AI model could predict wildfire risk based on vegetation cover or simulate how land cover might look under future climate or policy scenarios. Historic vegetation reconstructions (using early survey records or the 1850s Kansas land cover maps) could be used to compare then-and-now, enriching historical storytelling about the Kansas environment.

Disaster and Extreme Weather Data

- FEMA Disaster Declarations – Open data on all federally declared disasters in Kansas since 1953. FEMA's Disaster Declaration Summaries dataset lists every declared major disaster, emergency, or fire management assistance event, along with dates, affected

counties, and assistance programs . Examples include presidential disaster declarations for floods, tornadoes, severe storms, droughts, etc.

  - Coverage: Records from 1953 to present, updated continually. Each entry includes the disaster type (e.g. Tornado outbreak), incident date range, the counties included (note: county-level tagging is fully available post-1964) , and the declaration date. It also notes which federal assistance programs were activated (individual assistance, public assistance, etc.).

  - Access: FEMA provides this dataset on its website and via API (OpenFEMA). It can be downloaded as CSV/JSON in entirety or filtered via API queries (e.g., all disasters in Kansas, or all drought declarations nationwide). The data is also on Data.gov for bulk access.

  - License: Public domain (U.S. DHS/FEMA data). OpenFEMA datasets are intended for public use . The license is essentially government works free for reuse (FEMA just asks that users cite and be aware data is raw and may contain minor errors).

  - Potential Use: This is crucial for understanding the history of disasters in Kansas – from the 1951 Great Flood to Greensburg's 2007 tornado to recent wildfires. In the Frontier Matrix, one can layer disaster events over time to see patterns (e.g., increasing frequency of certain disaster types) or link them to other datasets (population or land use changes post-disaster). It also enables training models on the impacts of disasters – for instance, an AI could use past declaration data to predict which areas are most at risk or how long recovery takes, feeding into future resilience planning.

- NOAA Storm Events Database – A detailed record of severe weather occurrences compiled by the National Weather Service, including tornadoes, hail, thunderstorms, winter storms, and other extremes in Kansas. This database captures event specifics such as date, location (county or lat/long), magnitude (e.g. tornado EF rating, hail size), injuries/fatalities, and damage costs.

  - Coverage: 1950s to present for tornadoes (the database starts tornado records in 1950), and later start dates for other phenomena (hail, wind, etc. from 1955; floods and others by the 1990s). It is continuously updated by each NWS forecast office. Kansas data is comprehensive – e.g., every tornado that has touched down in Kansas is recorded, as well as major storms.

  - Access: NOAA's Storm Events query tool allows filtering by state, county, event type and downloading results in CSV . Bulk files by year are also downloadable. The data can be integrated into GIS by geocoding county-based events or using

provided coordinates for point-specific events.

- License: Public domain (U.S. Government). NOAA makes this data freely available for research and does not restrict usage.

- Potential Use: The Storm Events data brings granular disaster modeling capability. For example, one can map all tornado tracks in Kansas or analyze seasonality and trend (are winters becoming more extreme?). In the Frontier Matrix context, this could enrich historical timelines (e.g., marking the 1974 tornado outbreaks or 1993 Great Flood on a map alongside other data). It enables AI models to learn from past patterns – for instance, correlating soil moisture or temperature anomalies with subsequent disaster frequency, or predicting damage risks for different regions. Coupling storm data with FEMA declarations also distinguishes between event occurrence and its socio-economic impact (not every storm becomes a federal disaster, and that threshold itself is informative).

Historical Narratives and Oral Histories

- Kansas Memory (Kansas Historical Society Digital Archives) – A vast online repository of primary sources related to Kansas history. It contains nearly 600,000 items including photographs, letters, diaries, maps, newspaper clippings, and some audio recordings . While not a “dataset” in the numeric sense, it is a rich data source for textual and visual material that can be mined by AI or indexed alongside quantitative data.

    - Coverage: Spans the entirety of Kansas history – e.g., territorial-era letters, 19th-century Native American accounts, frontier town photographs, through mid-20th-century documents. Geographically covers communities across the state. There are also oral history transcripts and reminiscences in the collection (e.g., WPA era pioneer interviews).

    - Access: Web portal (kansasmemory.org) with search by keywords, topics, or collections. Items can be viewed and downloaded (images as JPEGs, documents as PDFs). Metadata (title, date, description, location) is provided for each item. For large-scale analysis, one might use the API or contact KHS for bulk data – but generally, scraping the site is required for big data uses.

    - License: Materials are made available for research and educational use. Many items are in the public domain (especially older than 1920s). KHS asks for attribution and may have usage restrictions on some copyrighted 20th-century materials. Overall, much of Kansas Memory’s content can be used freely for non-commercial purposes, and fees may apply for high-resolution or commercial

use.

    - Potential Use: Kansas Memory provides the human stories behind the data – useful for contextualizing and enriching the Frontier Matrix. For instance, one could associate a spike in 1870s immigration (from census data) with actual letters from immigrant families (from Kansas Memory) describing their journey. In AI terms, the text corpora in letters and newspapers could be used for natural language processing – training models to extract common themes (drought, war, prosperity) over decades. Image archives allow computer vision AI to learn from historical photographs (e.g., to identify farming scenes vs. urban scenes). By integrating these narratives, the platform moves beyond numbers, ensuring that voices of Kansans (including historically marginalized ones) are part of the modeled future.

- Kansas Oral History Project (KOHP) – A non-profit initiative that has recorded and transcribed interviews with notable Kansans, particularly focused on those involved in public policy and governance. This is a goldmine of firsthand accounts on late 20th-century Kansas politics, education, water policy, and more . Collections include interviews with former legislators, governors, community leaders, and diverse citizens.

    - Coverage: Approximately 170+ interviews (and growing). The focus is roughly 1960s–2010s in subject matter (e.g., participants recount civil rights in Kansas, school finance battles, rural development, etc.). While the emphasis is on governance, the content often delves into social and economic history as told by the interviewees.

    - Access: All interviews are available on the KOHP website (ksoralhistory.org) as streaming audio and text transcripts. They are organized into thematic collections ("Women Then and Now," "Kansas Governors," "Reflections on Water," etc.) . Transcripts can be read online or downloaded. KOHP also provides summaries and keywords for each interview.

    - License: The collected interviews (text and audio) are offered under a Creative Commons Attribution-NonCommercial-NoDerivs (CC BY-NC-ND 4.0) license . This means they can be freely used and shared for non-commercial purposes with attribution, but not altered or used commercially without permission.

    - Potential Use: These oral histories add a qualitative dimension to data-driven modeling. They allow the Frontier Matrix to incorporate insights like why certain policies succeeded or how communities felt during pivotal moments (e.g., the 1980s farm crisis from a legislator's perspective). In practical terms, one could use NLP on the transcripts to extract sentiment trends or key concerns over time (water rights, education funding, etc.), complementing the quantitative indicators. For AI enrichment, training language models on this Kansas-specific corpus

could enable more nuanced storytelling or Q&A about Kansas history. Moreover, integrating audio clips or quotes into the platform can make the past “speak” to users in a compelling way, ensuring that any future scenarios we model remain grounded in the lived experiences and voices of Kansans.

Sources: The compiled information above is drawn from a range of open data repositories and official resources, including GBIF , iNaturalist documentation , NOAA climate archives , Kansas Mesonet API docs , Cornell’s eBird data release notes , the Kansas GAP final report , U.S. Census Bureau datasets and NHGIS descriptions , USDA NASS QuickStats metadata , USDA Cropland Data Layer info , FEMA’s OpenFEMA data portal , NOAA Storm Events tool , Kansas Historic Resources Inventory site , NPS National Register data page , NRCS soil database description , NLCD land cover database documentation , and the Kansas Oral History Project and Kansas Memory archives . These sources are all open-access and provide the foundation for integrating Kansas’s past, present, and future into the Kansas Frontier Matrix platform.